

JavaScript Operators

What is an Operator?

An **Operator** is a symbol that performs an operation on one or more values (operands).

```
let sum = 10 + 5;

console.log(sum); // 15
```

Types of Operators

1. Arithmetic Operators
2. Assignment Operators
3. Comparison Operators
4. Logical Operators
5. Increment & Decrement Operators
6. String Operator
7. Ternary Operator
8. Type Operator

1. Arithmetic Operators

Used for mathematical calculations.

Operator	Meaning	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
%	Modulus	10 % 3
**	Exponent	2 ** 3

```
let a = 10;
let b = 3;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
console.log(a % b);
console.log(a ** b);
```

2. Assignment Operators

Used to assign or update values.

Operator	Example	Same As
=	<code>x = 5</code>	Assign value
+=	<code>x += 2</code>	<code>x = x + 2</code>
-=	<code>x -= 2</code>	<code>x = x - 2</code>
*=	<code>x *= 2</code>	<code>x = x * 2</code>
/=	<code>x /= 2</code>	<code>x = x / 2</code>
%=	<code>x %= 2</code>	<code>x = x % 2</code>

```
let x = 10;

x += 5;
console.log(x); // 15

x -= 3;
console.log(x); // 12
```

3. Comparison Operators

Used to compare two values.

Operator	Meaning	Example
==	Equal (value only)	<code>5 == "5"</code>
===	Strict Equal	<code>5 === "5"</code>
!=	Not Equal	<code>5 != 4</code>
!==	Strict Not Equal	<code>5 !== "5"</code>
>	Greater Than	<code>10 > 5</code>
<	Less Than	<code>5 < 10</code>
>=	Greater Than or Equal	<code>5 >= 5</code>
<=	Less Than or Equal	<code>5 <= 10</code>

```
console.log(10 > 5);
console.log(10 == "10");
console.log(10 === "10");
```

4. Logical Operators

Used to combine conditions.

Operator	Meaning
&&	AND
!	NOT

```
let age = 20;
let hasID = true;

console.log(age >= 18 && hasID);
console.log(age < 18 || hasID);
console.log(!hasID);
```

5. Increment & Decrement Operators

Increase or decrease a value by 1.

Operator	Meaning
++	Increment
--	Decrement

```
let count = 5;

count++;
console.log(count); // 6

count--;
console.log(count); // 5
```

6. String Operator

The `+` operator joins strings.

```
let firstName = "MD";
let lastName = "Rokonuzzaman";

console.log(firstName + " " + lastName);
```

Output:

```
MD Rokonzaman
```

| 7. Ternary Operator

A short way to write an `if...else`.

Syntax

```
condition ? trueValue : falseValue;
```

```
let age = 20;

let result = age >= 18 ? "Adult" : "Minor";

console.log(result);
```

| 8. Type Operator

The `typeof` operator returns the data type of a value.

```
console.log(typeof "Hello");
console.log(typeof 100);
console.log(typeof true);
console.log(typeof []);
console.log(typeof {});
```

| Operator Precedence

Operators with higher precedence execute first.

```
let result = 10 + 5 * 2;

console.log(result);
```

Output:

```
20
```

Because `*` has higher precedence than `+`.

Summary

Category	Purpose
Arithmetic	Mathematical calculations
Assignment	Assign or update values
Comparison	Compare values
Logical	Combine conditions
Increment/Decrement	Increase or decrease by 1
String	Join strings
Ternary	Short <code>if...else</code>
Type	Check data type

Easy Way to Remember

- `+ - * / % **` → Arithmetic
- `= += -= *= /= %=` → Assignment
- `== === != > < >= <=` → Comparison
- `&& || !` → Logical
- `++ --` → Increment / Decrement
- `+` → String Concatenation
- `? :` → Ternary
- `typeof` → Type Checking

Interview Questions

Q1. What is the difference between `==` and `===`?

- `==` compares **only values** (type conversion occurs).
- `===` compares **both value and data type**.

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

Q2. What is the difference between `&&` and `||`?

- `&&` (AND): **All conditions must be true.**
- `||` (OR): **At least one condition must be true.**

```
console.log(true && false); // false  
console.log(true || false); // true
```